



2012

**TRIAL
HIGHER SCHOOL CERTIFICATE**

MATHEMATICS EXTENSION 1

General Instructions:

- Reading Time - 5 minutes

Working time - 2 hours

- Write using black or blue pen.
- Board - approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question

Total marks – 70

Section 1 10 marks

Attempt Questions 1 – 10

Section II 60 Marks

Attempt Questions 11 – 14

All questions are of equal value

SECTION 1 **Multiple Choice Questions:** **Circle the correct answer**

1. The polynomial $P(x) = x^4 - kx^3 - 2x + 33$ has $(x - 3)$ as a factor.

What is the value of k ?

(A) -5

(B) -4

(C) 4

(D) 5

2. The velocity of a particle moving along the x axis is given by $v^2 = 24 + 2x - x^2$. Which of the following expressions is the correct equation for the acceleration of the particle in terms of x ?

(A) $1 - x$

(B) $1 - 2x$

(C) $12x + \frac{x^2}{2} - \frac{x^3}{6}$

(D) $24x + x^2 - \frac{x^3}{3}$

3. If $f(x) = e^{x+2}$ what is the inverse function $f^{-1}(x)$?

(A) $f^{-1}(x) = e^{y-2}$

(B) $f^{-1}(x) = e^{y+2}$

(C) $f^{-1}(x) = \log_e x - 2$

(D) $f^{-1}(x) = \log_e x + 2$

4. At a football club a team of 11 players is to be chosen from a pool of 30 players consisting of 18 Australian-born players and 12 players born elsewhere. What is the probability that the team will consist of all Australian-born players?

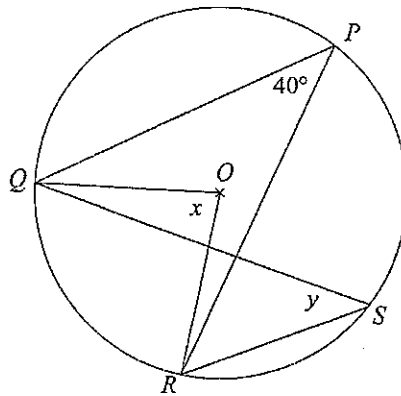
(A) $\frac{{}^{18}C_{11}}{{}^{30}C_{11}}$

(B) $\frac{{}^{30}C_{11}}{{}^{18}C_{11}}$

(C) $\frac{{}^{18}C_{12}}{{}^{30}C_{12}}$

(D) $\frac{{}^{30}C_{12}}{{}^{18}C_{12}}$

5. P , Q , R and S are points on a circle with centre O . $\angle QPR = 40^\circ$.



Why are the values of x and y ?

(A) $x = 40^\circ$ and $y = 20^\circ$

(B) $x = 40^\circ$ and $y = 40^\circ$

(C) $x = 80^\circ$ and $y = 20^\circ$

(D) $x = 80^\circ$ and $y = 40^\circ$

6. A curve has parametric equations $x = \frac{2}{t}$ and $y = 2t^2$.

What is Cartesian equation of this curve?

(A) $y = \frac{4}{x}$

(B) $y = \frac{8}{x}$

(C) $y = \frac{4}{x^2}$

(D) $y = \frac{8}{x^2}$

7. What is the solution to the equation $|2x - 5| = -3x$?

- (A) $x = -5$
- (B) $x = -1$
- (C) $x = 1$
- (D) $x = 5$

8. What is the exact value of $\tan 75^\circ$?

- (A) $2 - \sqrt{3}$
- (B) $4 - \sqrt{3}$
- (C) $2 + \sqrt{3}$
- (D) $4 + \sqrt{3}$

9. A parabola has the parametric equations $x = 12t$ and $y = -6t^2$.

What are the coordinates of the focus?

- (A) $(-6, 0)$
- (B) $(0, -6)$
- (C) $(6, 0)$
- (D) $(0, 6)$

10. How many four-digit numbers can be formed with the digits 1, 2, 3, 4 and 5 if no digit is repeated?

- (A) 20
- (B) 120
- (C) 625
- (D) 3125

SECTION II

Question 11. (15 marks)

a) Solve $\frac{x}{x-2} \geq 2$ 2

b) Find the coordinates of the point P which divides the interval AB *externally* in the ratio 1:3, given A = (1,4) and B = (5,2) 2

c) Evaluate

i) $\int_1^2 \frac{1}{\sqrt{4-x^2}} dx$ 2

ii) $\int_{-1}^0 x\sqrt{1+x} dx$, using the substitution $u = 1+x$ 2

d) Find the acute angle between the lines $y = 2x - 1$ and $y = \frac{1}{3}x + 1$ 2

e) Find the number of ways in which two consonants and three vowels can be chosen from the letters of the word EQUATION? 1

f) The polynomial $P(x) = x^3 + px^2 + qx + r$ has real roots

$\sqrt{k}$, $-\sqrt{k}$, and α .

i) Explain why $\alpha + p = 0$ 1

ii) Show that $k\alpha = r$ 1

iii) Show that $pq = r$ 2

Question 12. (15 marks)

- a) Find $\int \sin^2 3x dx$ **2**
- b) Simplify $\sin 2\theta(\tan \theta + \cot \theta)$ **2**
- c) Consider the function $f(x) = 3 \cos^{-1}\left(\frac{x}{2}\right)$
- i) Sketch the graph $y = f(x)$ **3**
- ii) Find the gradient of the tangent to the curve at the point
where $x = \sqrt{3}$ **1**
- d) If $\alpha = \sin^{-1}\left(\frac{8}{17}\right)$ and $\beta = \tan^{-1}\left(\frac{3}{4}\right)$, calculate the exact value of $\sin(\alpha - \beta)$ **3**
- e) Solve $\cos 2\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$ **2**
- f) Differentiate $e^x \cos^{-1} x$ **2**

Question 13. (15 marks)

- a)) Prove by mathematical induction that
- $$1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + n(n+1)^2 = \frac{1}{12}n(n+1)(n+2)(3n+5)$$
- 4**
- b)) Let $g(x) = 2x^3 + x + 4$
- i) Show that $g(x) = 0$ has a root between integers -1 and -2 **1**
- ii) Taking $x = -1.5$ as the first approximation to this root, use one application
of Newton's method to obtain a better approximation for this root. **2**

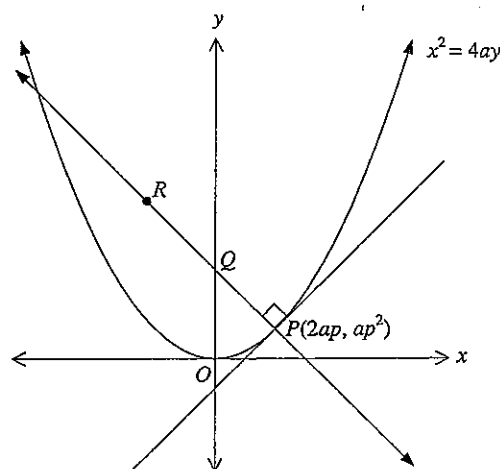
Question 13 continued

c) The velocity, $vm s^{-1}$, of a particle moving in simple harmonic motion along the x-axis is given by $v^2 = 8 - 2x - x^2$, where x is in metres.

- i) Between which two points is the particle oscillating? 1
- ii) Find the centre of the motion. 1
- iii) Find the maximum speed. 1
- iv) Find an expression for the acceleration of the particle in terms of x . 2

- d) i) Express $\cos x - \sin x$ in the form $A \cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$ 1
- ii) Hence, or otherwise, solve $\cos x - \sin x = 1$ for $0 \leq x \leq 2\pi$ 2

Question 14. 15 marks)



a) The diagram shows a variable point $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$.

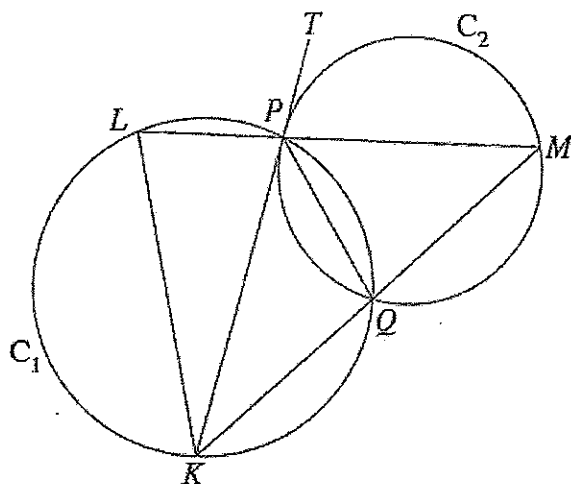
The normal to the parabola at P intersects the y axis at Q . The point Q is the mid point of PR .

The equation of the normal is $x + py - 2ap - ap^3 = 0$. (Do Not prove this.)

- i) Find the coordinates of the point Q . 1
- ii) The locus of the point R is a parabola. Find the equation of this parabola in Cartesian form and state its vertex. 2

Question 14 continued

b)



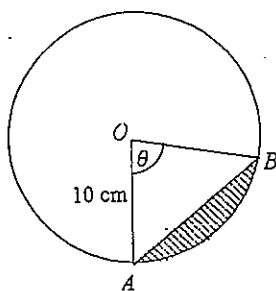
Two circles C_1 and C_2 intersect at P and Q as shown in the diagram. The tangent TP to C_2 at P meets C_1 at K . The line KQ meets C_2 at M . The line MP meets C_1 at L .

Copy the diagram into your writing booklet.

Prove that $\triangle PKL$ is isosceles.

3

c)



A circle has centre O and radius 10 cm . OA is a fixed radius of the circle. OB is a variable radius which moves so that $\angle AOB = \theta$ is increasing at a constant rate of 0.01 radians per second. The minor segment of the circle cut off by the chord AB has area $S\text{ cm}^2$

Find the rate at which S is increasing when $\theta = \frac{\pi}{3}$.

2

Question 14 continued.

d) Molten metal at a temperature of 1400°C is poured into moulds to

form machine parts. After 15 minutes the metal has cooled to 995°C .

If the temperature of the surroundings is 35°C , then the rate of cooling is approximately given by;

$$\frac{dT}{dt} = -k(T - 35) \text{ where } k \text{ is a positive constant.}$$

Show that a solution of this equation is $T = 35 + Ae^{-kt}$

where A is a constant.

1

i) Find the value of k , correct to three decimal places.

1

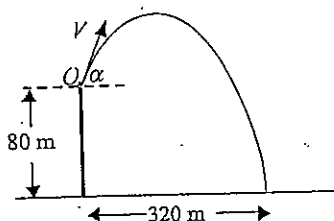
iii) The metal can be taken out of the moulds when its temperature has

dropped to 200°C . How long after the metal has been poured will

this temperature be reached? (answer correct to the nearest minute.)

2

e)



A particle is projected with speed $V\text{ms}^{-1}$ at an angle α above the horizontal from a point O at the edge of a vertical cliff which is 80m above horizontal ground. The particle moves in a vertical plane under gravity where the acceleration due to gravity is 10ms^{-2} . It reaches its greatest height after 3 seconds and hits the ground at a horizontal distance 320m from the foot of the cliff.

The horizontal and vertical displacements, x and y metres respectively, of the particle from the point O after t seconds are given by $x = Vt\cos\alpha$ and $y = -5t^2 + Vt\sin\alpha$. (Do not prove this)

i) Show that $V\sin\alpha = 30$

1

ii) Show that the particle hits the ground after 8 seconds.

2

Section I

1. C
2. A
3. C
4. A
5. D
6. D
7. A
8. C
9. B
10. B

Section II

Question 11

a) $\frac{x}{x-2} \geq 2 \quad x \neq 2$

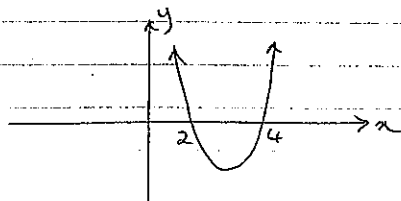
$$\frac{x}{x-2} \geq 2 \Rightarrow x(x-2) \geq 2(x-2)^2$$

$$x(x-2) \geq 2(x-2)^2$$

$$0 \geq 2(x-2)^2 - x(x-2)$$

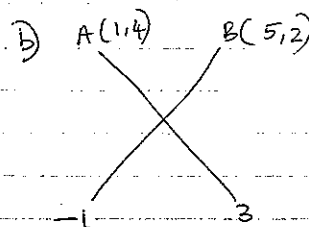
$$0 \geq (x-2)(2x-4-x)$$

$$0 \geq (x-2)(x-4) \quad (1)$$



$$2 \leq x \leq 4 \quad (1)$$

Question 11 continued



$$x = \frac{3-5}{2} = -1$$

$$y = \frac{12-2}{2} = 5$$

$$\therefore P = (-1, 5) \quad (2)$$

c) (i) $\int_1^2 \frac{1}{\sqrt{4-x^2}} dx = \left[\sin^{-1} \frac{x}{2} \right]_1^2 = \sin^{-1} 1 - \sin^{-1} \frac{1}{2}$
 $= \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3} \quad (2)$

(ii) $\int_{-1}^0 x\sqrt{1+x} dx$ $\begin{array}{l|l} u=1+x & \text{limits} \\ du=dx & \text{if } x=-1, u=0 \\ & \text{if } x=0, u=1 \end{array}$

$$= \int_0^1 (u-1)u^{\frac{1}{2}} du$$

$$= \int_0^1 (u^{\frac{3}{2}} - u^{\frac{1}{2}}) du = \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_0^1$$

$$= \left(\frac{2}{5} - \frac{2}{3} \right) - (0-0)$$

$$= -\frac{4}{15} \quad (2)$$

d) $y = 2x-1$ $y = \frac{1}{3}x+1$
 $m_1 = 2$ $m_2 = \frac{1}{3}$

$$\tan \alpha = \left| \frac{2 - \frac{1}{3}}{1 + \frac{2}{3}} \right| = 1 \quad \therefore \alpha = 45^\circ \quad (2)$$

e) vowels = 5 ; consonants = 3
 $\therefore {}^3C_2 \times {}^5C_3 = 30 \text{ ways} \quad (1)$

Question 11 Continued:

f) $P(x) = x^3 + px^2 + qx + r$

(i) Let the roots be x_1, x_2 and x_3
 $x_1 = \sqrt{k}, x_2 = -\sqrt{k}, x_3 = \alpha$

$$x_1 + x_2 + x_3 = \frac{-b}{a} = -p$$

$$\therefore \sqrt{k} - \sqrt{k} + \alpha = -p \Rightarrow \alpha + p = 0 \quad (1)$$

(ii) $x_1 x_2 x_3 = -r$

$$(\sqrt{k})(-\sqrt{k})\alpha = -r$$

$$-k\alpha = -r \Rightarrow k\alpha = r \quad (1)$$

(iii) $x_1 x_2 + x_2 x_3 + x_1 x_3 = q$

$$(\sqrt{k})(-\sqrt{k}) + (-\sqrt{k})\alpha + (\sqrt{k})\alpha = q$$

$$-k - \sqrt{k}\alpha + \sqrt{k}\alpha = q \Rightarrow -k = q$$

Also $p = -q$ from (i)

$$\therefore pq = (-k)(-q) = kq = r \quad (2)$$

Question 12

a) $\cos 6x = 1 - 2\sin^2 3x$

$$\sin^2 3x = \frac{1 - \cos 6x}{2} \quad (1)$$

$$\therefore \int \sin^2 3x dx = \frac{1}{2} \int (1 - \cos 6x) dx$$

$$= \frac{1}{2} \left(x - \frac{\sin 6x}{6} \right) + C \quad (1)$$

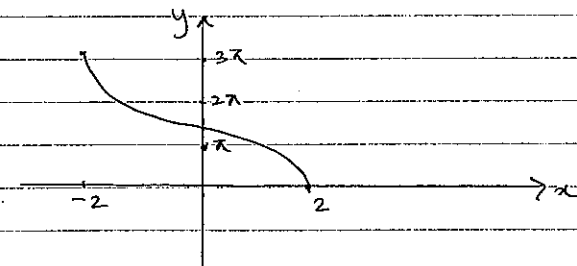
b) $\sin 2\theta (\tan \theta + \cot \theta) = 2 \sin \theta \cos \theta \left(\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)$

$$= 2 \sin^2 \theta + 2 \cos^2 \theta = 2 \quad (2)$$

c) (i) $f(x) = 3 \cos^{-1} \left(\frac{x}{2} \right)$

$$D: -1 \leq \frac{x}{2} \leq 1 \Rightarrow -2 \leq x \leq 2 \quad (1)$$

$$R: 0 \leq f(x) < 3\pi \quad (1)$$



(ii) $f(x) = 3 \cos^{-1} \frac{x}{2} \quad \therefore f'(x) = 3 \cdot \frac{-1}{\sqrt{1 - \frac{x^2}{4}}} \cdot \frac{1}{2}$

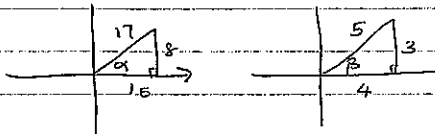
$$= \frac{-3}{\sqrt{4 - x^2}}$$

at $x = \sqrt{3}, f'(x) = \frac{-3}{\sqrt{4 - 3}} = -3 \quad (1)$

Question 12 continued

d) $\alpha = \sin^{-1}\left(\frac{8}{17}\right) \Rightarrow \sin \alpha = \frac{8}{17}$

$\beta = \tan^{-1}\left(\frac{3}{4}\right) \Rightarrow \tan \beta = \frac{3}{4}$



①

$$\begin{aligned} \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ &= \frac{8}{17} \cdot \frac{4}{5} - \frac{15}{17} \cdot \frac{3}{5} = \frac{-13}{15} \end{aligned} \quad \text{②}$$

e) $\cos 2\theta = \cos \theta$ for $0 \leq \theta < 2\pi$.

$$2\cos^2 \theta - 1 = \cos \theta$$

$$2\cos^2 \theta - \cos \theta - 1 = 0 \Rightarrow (2\cos \theta + 1)(\cos \theta - 1) = 0$$

$$\therefore \cos \theta = -\frac{1}{2} \text{ or } \cos \theta = 1 \quad \text{①}$$

$$\therefore \theta = \frac{2\pi}{3}, \frac{4\pi}{3}, 0, 2\pi \therefore = 0, \frac{2\pi}{3}, \frac{4\pi}{3}, 2\pi \quad \text{①}$$

f) $y = e^x \cos^{-1} x$

$u = e^x$	$v = \cos^{-1} x$
$u' = e^x$	$v' = \frac{-1}{\sqrt{1-x^2}}$

$$\begin{aligned} \therefore y' &= vu' + uv' \\ &= \cos^{-1} x \cdot e^x + e^x \frac{(-1)}{\sqrt{1-x^2}} \\ &= e^x \left[\cos^{-1} x - \frac{1}{\sqrt{1-x^2}} \right] \quad \text{②} \end{aligned}$$

Question 13

a) $1 \times 2^2 + 2 \times 3^2 + \dots + n(n+1)^2 = \frac{1}{12} n(n+1)(n+2)(3n+5)$

• prove true for $n=1$

LHS $= 1 \times 2^2 = 4$

RHS $= \frac{1}{12} (2)(3)(4) = 4$

$\therefore$ true for $n=1$

• assume true for $n=k$

i.e. $1 \times 2^2 + 2 \times 3^2 + \dots + k(k+1)^2 = \frac{1}{12} k(k+1)(k+2)(3k+5)$

• prove true for $n=k+1$

b) (i) $1, 2, 0$

(ii) $1, 2, 2$

(iii) $1, 2, 2$

(iv) $1, 2, 1$

(v) $1, 2, 1$

(vi) $1, 2, 1$

(vii) $1, 2, 1$

d) $A = \sqrt{2} \times \frac{\pi}{4} = \frac{1}{12} (k+1)(k+2) [k(3k+5) + 12(k+2)]$

$= \frac{1}{12} (k+1)(k+2) [3k^2 + 17k + 24]$

$= \frac{1}{12} (k+1)(k+2)(k+3)(3k+8) = \text{RHS}$

If true for $n=k$, then also true for $n=k+1$. But it is true for $n=1$

$\therefore$ Also true for $n=2, 3, \dots, 2$

$\therefore$ By induction, true for all positive integers.

Question 13 continued

b) (i) $g(x) = 2x^3 + x + 4$

$$g(-1) = 2(-1)^3 + (-1) + 4 = 1 > 0$$

$$f(-2) = 2(-2)^3 + (-2) + 4 = -14 < 0$$

Since the sign changes and the curve is continuous, there is a root between $x = -1$ and $x = -2$

(ii) $g(x) = 2x^3 + x + 4$

$$g'(x) = 6x^2 + 1$$

$$x_1 = -1.5, \quad g(x_1) = 2(-1.5)^3 + (-1.5) + 4 = -4.25$$

$$g'(x_1) = 6(-1.5)^2 + 1 = 14.5$$

$$\therefore x_2 = x_1 - \frac{g(x_1)}{g'(x_1)} = -1.5 - \frac{-4.25}{14.5} = -1.2$$

c) (i) $v^2 = 8 - 2x - x^2$

At the end points $v = 0$

$$8 - 2x - x^2 = 0 \Rightarrow (4+x)(2-x) = 0$$

$$\therefore x = -4 \text{ or } x = 2$$

$\therefore$ the particle is oscillating between $x = -4$ and $x = 2$

(ii) $-4 \quad \text{---} \quad 2$
Centre of motion: $x = \frac{-4+2}{2}$

$$x = -1$$

Question 13 continued

c (iii) The maximum speed occurs at the centre of motion.

$$\therefore v^2 = 8 - 2(-1) - (-1)^2 = 8 + 2 - 1 = 9$$

$$\therefore \text{max. speed} = 3 \text{ m/s}$$

(iv) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$= \frac{d}{dx} \left(4 - x - \frac{x^2}{2} \right) = -1 - x$$

d) (i) $\cos x - \sin x = A \cos(x + \alpha)$ $0 < \alpha < \frac{\pi}{2}$

$$\cos x - \sin x = A \cos x \cos \alpha - A \sin x \sin \alpha$$

$$A \cos \alpha = 1; \quad A \sin \alpha = 1$$

$$\therefore A = \sqrt{2}; \quad \alpha = \frac{\pi}{4}$$

$$\therefore \cos x - \sin x = \sqrt{2} \cos \left(x + \frac{\pi}{4} \right)$$

(ii) $\sqrt{2} \cos \left(x + \frac{\pi}{4} \right) = 1$ $0 \leq x < 2\pi$

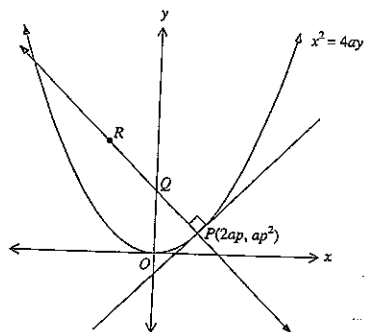
$$\frac{\pi}{4} \leq x + \frac{\pi}{4} < \frac{9\pi}{4}$$

$$\therefore x + \frac{\pi}{4} = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}$$

$$\therefore x = 0, \frac{3\pi}{2}, 2\pi$$

Question 14

a)



(i) The eqⁿ of normal: $x + py - 2ap - ap^3 = 0$

intersects the y axis at Q. $\therefore$ sub in $x=0$.

$$0 + py - 2ap - ap^3 = 0$$

$$\therefore y = \frac{2ap + ap^3}{p} = 2a + ap^2$$

$$\therefore Q = (0, 2a + ap^2) \quad (1)$$

(ii) Let $R = (x, y)$, Q is the mid point of PR

$$\therefore \frac{x + 2ap}{2} = 0 \Rightarrow x = -2ap \quad (1)$$

$$\frac{y + ap^2}{2} = 2a + ap^2 \Rightarrow y = 4a + ap^2 \quad (2)$$

eliminate p from (i) and (ii)

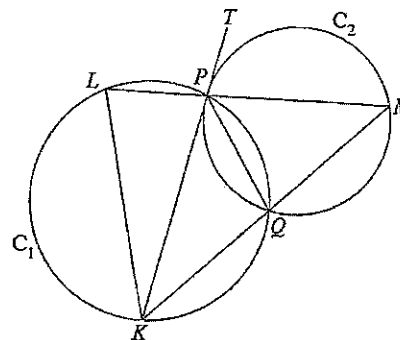
$$y = 4a + a \left(\frac{-x}{2a} \right)^2 = 4a + \frac{x^2}{4a}$$

$$\therefore x^2 = 4a(y - 4a) \quad (2)$$

$$\therefore \text{Vertex} = (0, 4a)$$

Question 14 continued

b)



$$\angle TPM = \angle LPK \quad (\text{vertically opposite } \angle \text{ s are equal})$$

$$\angle TPM = \angle PQM \quad (\angle \text{ in the alternate segment in } C_2 \text{ with tangent TP})$$

$$\therefore \angle LPK = \angle PQM \quad (1)$$

$$\angle PQM = \angle PLK \quad (\text{exterior } \angle \text{ to a cyclic quadrilateral in } C_1 = \text{opposite interior } \angle)$$

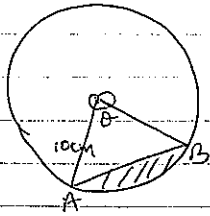
(1)

$$\therefore \angle LPK = \angle PLK \Rightarrow PK = LK$$

$$\therefore \triangle PKL \text{ is isosceles (sides opposite equal } \angle \text{ s)}$$

Question 14 continued

c)



$$S = \frac{1}{2} \times 10^2 (\theta - \sin \theta) = 50(\theta - \sin \theta)$$

$$\therefore \frac{ds}{d\theta} = 50(1 - \cos \theta)$$

$$\frac{d\theta}{dt} = 0.01 \text{ rad/sec}$$

$$\frac{ds}{dt} = \frac{ds}{d\theta} \times \frac{d\theta}{dt}$$

$$\therefore \frac{ds}{dt} = 50(1 - \cos \theta) \times 0.01$$

$$\text{when } \theta = \frac{\pi}{3}, \frac{ds}{dt} = 50(1 - \frac{1}{2}) \times 0.01 = 0.25$$

Area increases at $0.25 \text{ cm}^2/\text{s}$

d) (i) $T = 35 + Ae^{-kt}$
 $\frac{dT}{dt} = -k \cdot Ae^{-kt} = -k(T - 35)$ (1)

(ii) when $t=0$, $T=1400$

$$\therefore 1400 = 35 + A \Rightarrow A = 1365$$

when $t=15$, $T=995$

$$\therefore 995 = 35 + 1365e^{-15k} \Rightarrow e^{-15k} = \frac{960}{1365}$$

$$\therefore -15k = \log \frac{960}{1365}$$

$$\therefore k = -\frac{1}{15} \log \frac{960}{1365} = 0.02346 \dots = 0.023$$

Question 14 continued

d) (i) when $T=200$,
 $200 = 35 + 1365e^{-kt}$

$$\frac{-kt}{e} = \frac{165}{1365}$$

$$\therefore -kt = \log \frac{165}{1365} = \log \frac{11}{91}$$

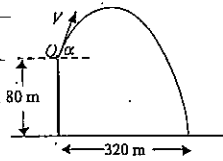
$$\therefore t = -\frac{\log \frac{11}{91}}{k} = 90.047 \dots = 90 \text{ minutes}$$

e) (i) $y = -5t^2 + vt \sin \alpha$

$$\dot{y} = -10t + v \sin \alpha$$

when $t=3$, $\dot{y}=0 \therefore 0 = -30 + v \sin \alpha$

$$\therefore v \sin \alpha = 30$$
 (1)



(ii) when it hits the ground, $y = -80$

$$\therefore -80 = -5t^2 + 30t$$

$$\therefore t^2 - 6t - 16 = 0$$

$$(t-8)(t+2) = 0 \therefore t = -2 \text{ or } 8$$

as $t \geq 0$, $t = 8$. (2)